

Homework 6

Instructions

Complete the following problems and submit files and code to Gradescope. Your work will be manually graded for this assignment, so there are no strict code formatting requirements.

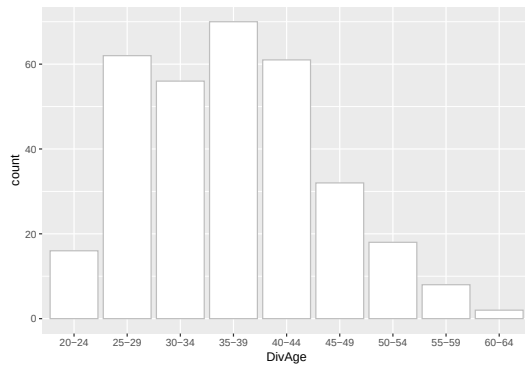
Problem 1

Recall the data file `nym2021.txt` that contains information regarding a sample of finishers from the 2021 New York City Marathon.

a

Plot the number of finishers in each age division as white bars with a gray outline.

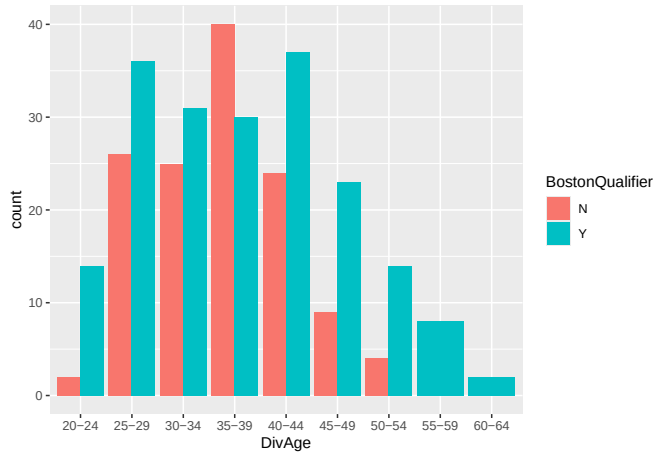
Here is an example. Do not worry about creating this image exactly—you will not be held accountable for anything not requested in textual instructions.



b

Separate/group the bars of the plot created in part a by whether or not the finishers qualified for the Boston Marathon, such that each qualifying category is represented by a different color. The bars for each qualifying category should be positioned side-by-side.

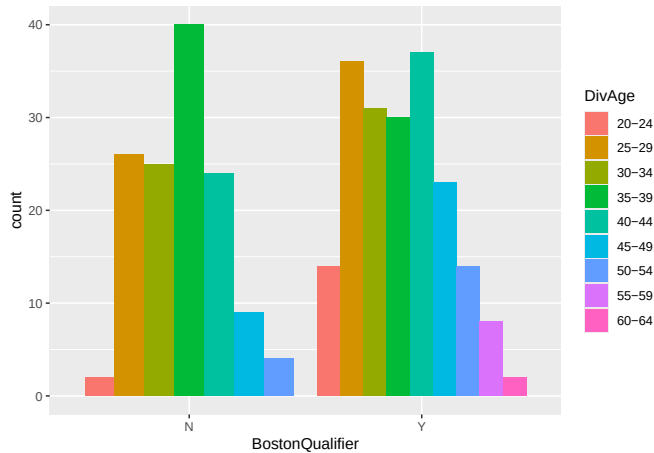
Here is an example. Do not worry about creating this image exactly—you will not be held accountable for anything not requested in textual instructions.



c

Modify the display of information in part b such that the Boston Marathon qualifying categories are separated by age division. Each age division should be represented by a different color. The bars for each age division should be positioned side-by-side.

Here is an example. Do not worry about creating this image exactly—you will not be held accountable for anything not requested in textual instructions.



d

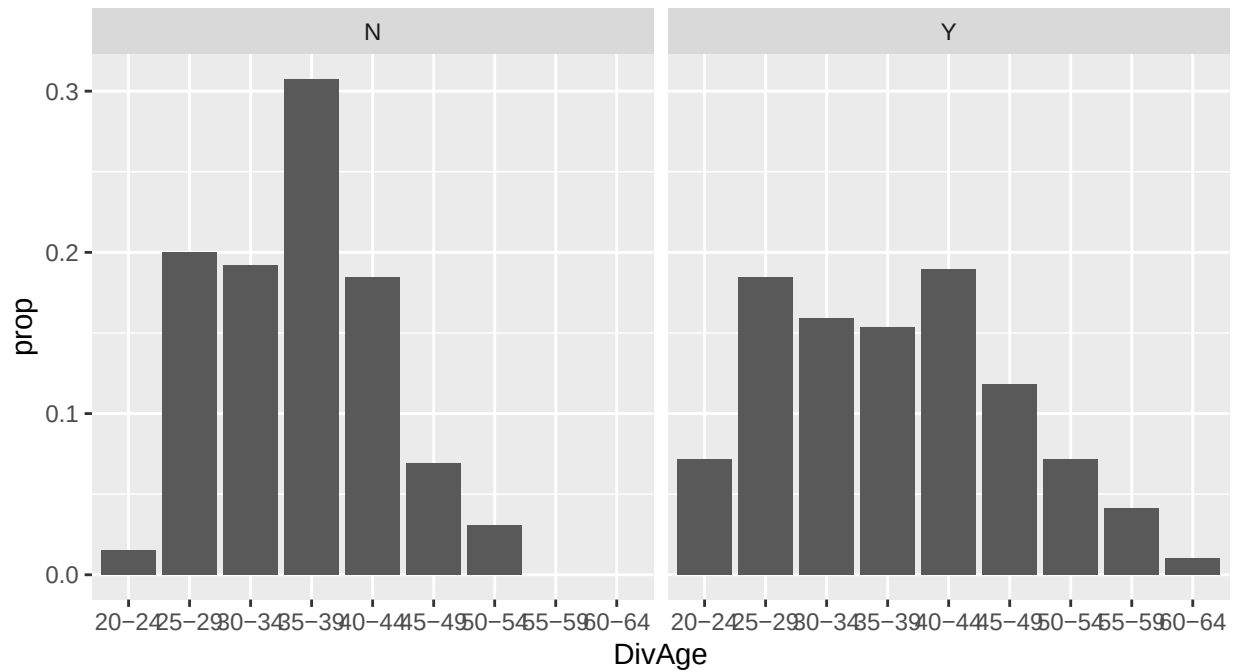
What conclusions can you draw from the plots created in parts b and c?

Type your response in the appropriate Gradescope free response box.

e

You will reproduce the graphic shown below in a later problem. It shows the proportion of finishers of each Boston Marathon qualifying category across the age divisions relative to the overall number of finishers of each qualifying category.

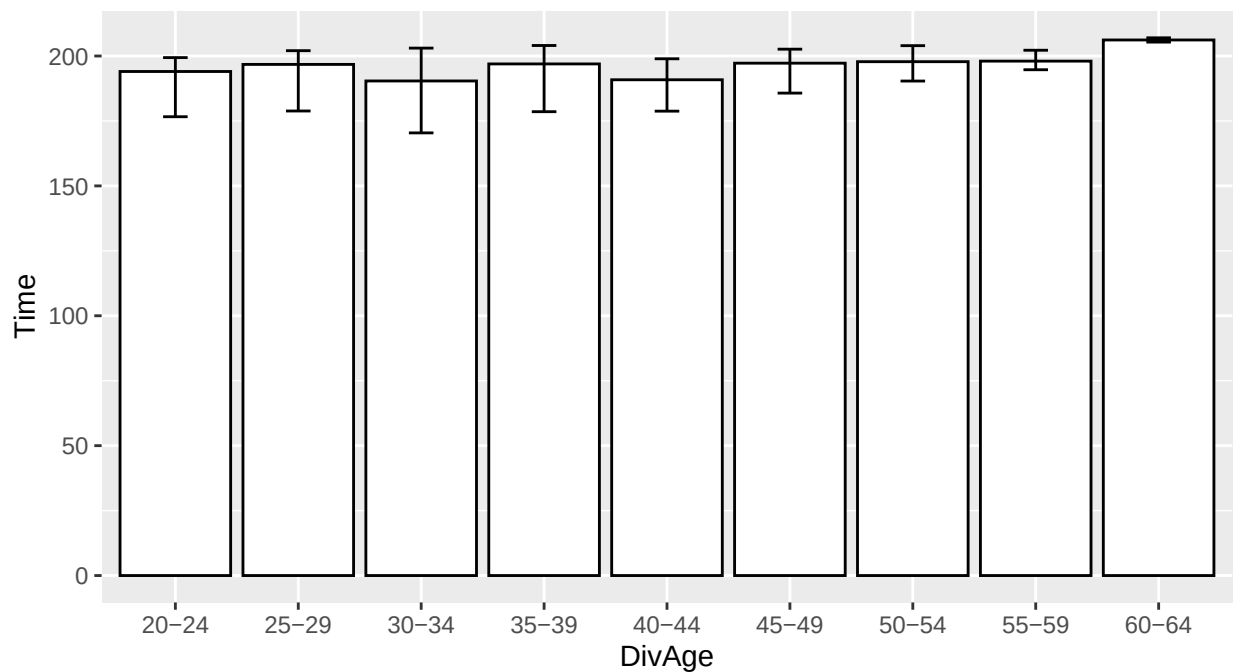
What conclusions can you draw from this plot? Does this plot change your conclusions drawn in part d? Type your response in the appropriate Gradescope free response box.



f

Create a bar graph for the median finishing time of each age division. The bars should be white with a black outline. Add error bars that represent the interquartile range (Q1 to Q3) and reduce their width to 20% of the width of the bars.

Here is an example. Do not worry about creating this image exactly—you will not be held accountable for anything not requested in textual instructions.



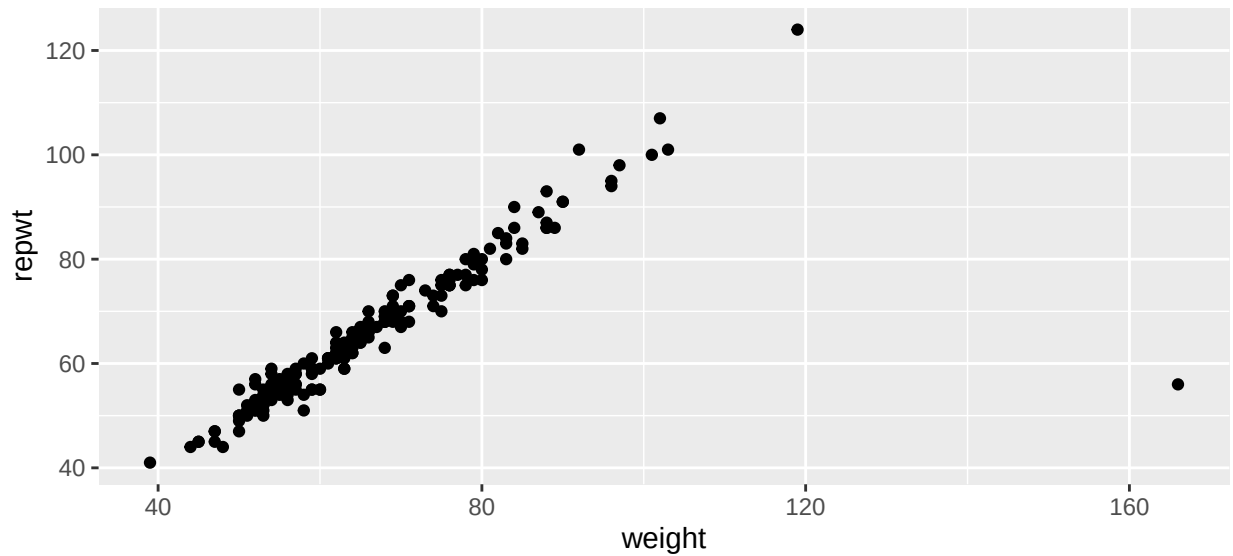
g

What conclusions can you draw about finishing times across age divisions from the plot created in part f? Type your response in the appropriate Gradescope free response box.

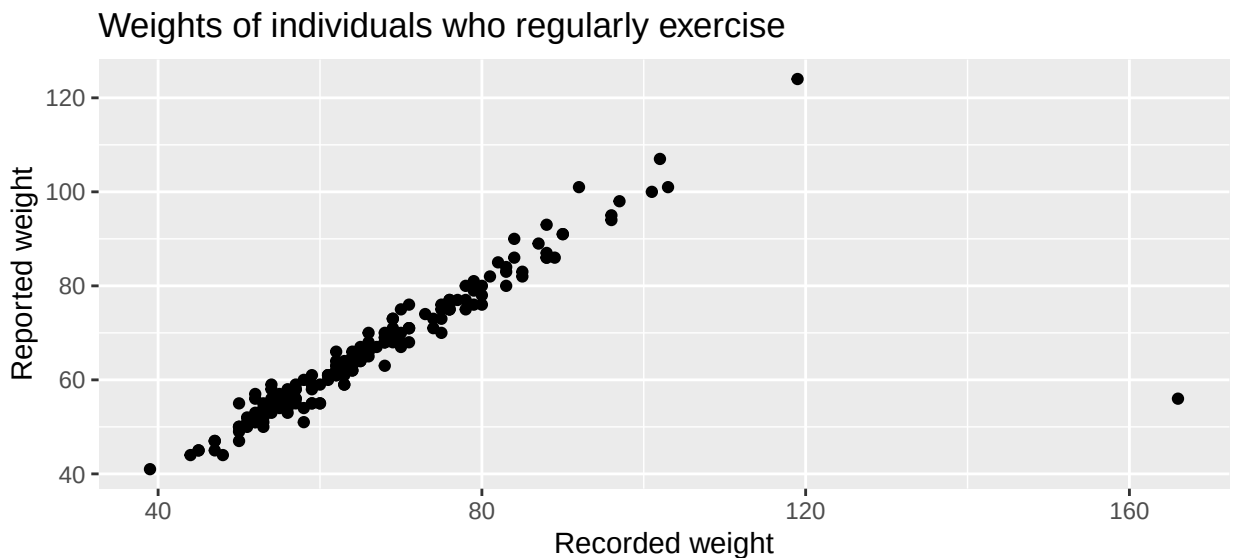
For each of the following problems, you should first recreate the graphic identified and save it as **GraphicX**, where X is the problem number. If your original graphic was not fully correct, you should modify the code that you use to create **GraphicX**, so that you start with the correct graphic as shown.

Next, add layers to **GraphicX** to create the *modified* graphic.

Class example:



Modified:



```
Graphic0 <- ggplot(Davis, aes(x=weight, y=repwt)) + geom_point()
Graphic0 + labs(title="Weights of individuals who regularly exercise",
               x="Recorded weight", y="Reported weight")
```

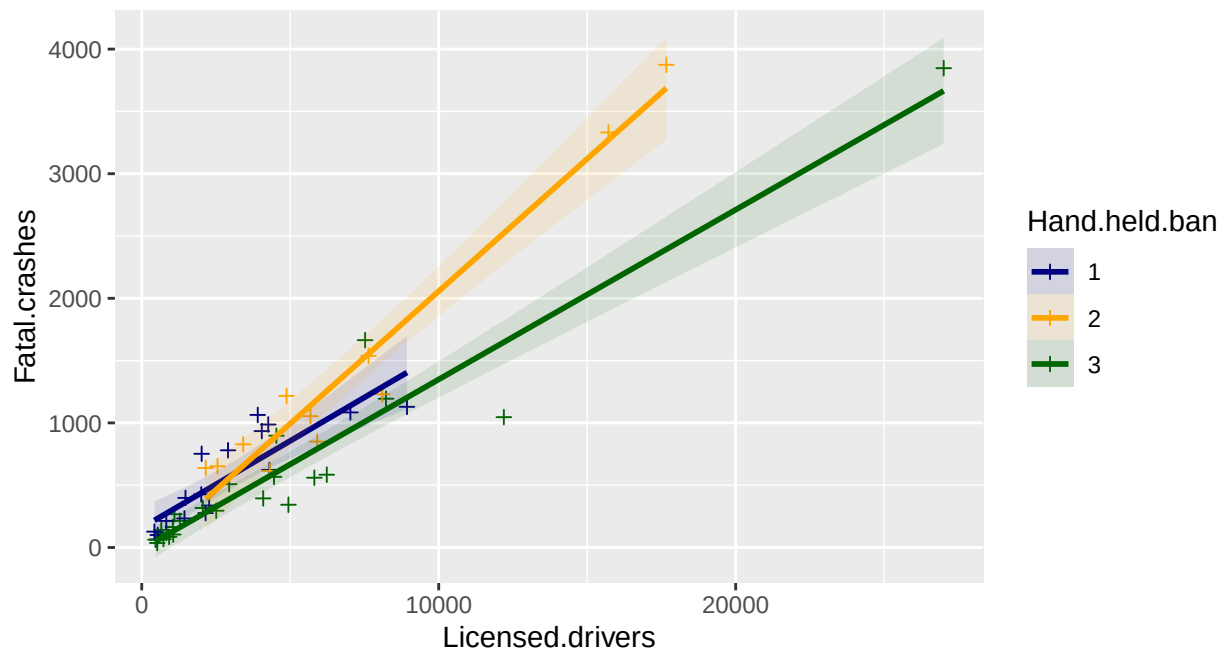
NB1: **You may need to reuse select layers from GraphicX** to conform to the code structure required here; outside of this structure you would modify the layer in **GraphicX** instead.

NB2: Print **ONLY** the modified graphic (and the code used to create it).

NB3: be careful about copy/pasting code from this pdf. Many of you have reported that the encoding gets messed up on certain characters when you do this.

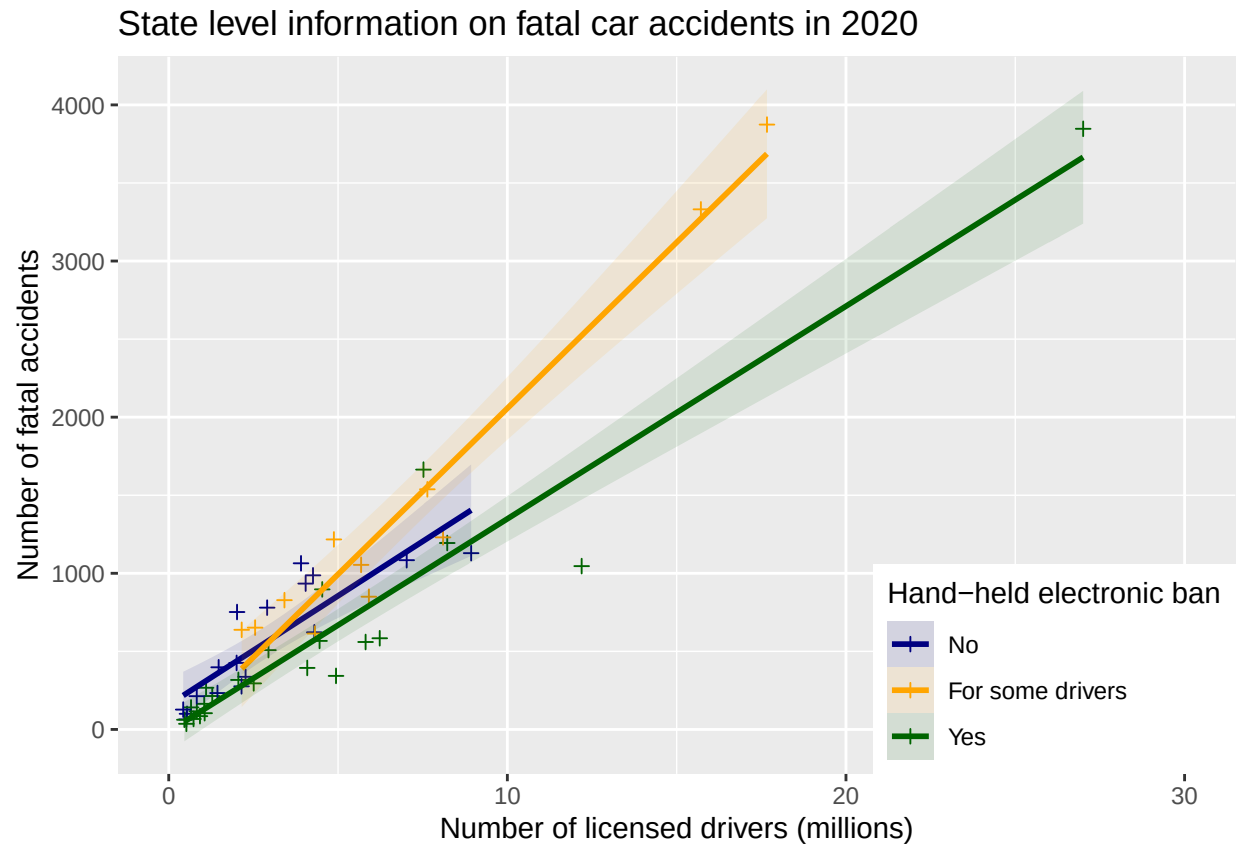
Problem 2

```
crash.df <- read.csv("state_crashes.csv",
                    header=TRUE)
crash.df$Hand.held.ban <- as.factor(crash.df$Hand.held.ban)
ggplot(crash.df, aes(x=Licensed.drivers, y=Fatal.crashes,
                    color=Hand.held.ban, fill=Hand.held.ban)) +
  geom_point(shape=3) +
  scale_color_manual("Hand.held.ban",
                    values=c("1"="navyblue", "2"="orange", "3"="darkgreen")) +
  geom_smooth(method=lm, alpha=0.1) +
  scale_fill_manual("Hand.held.ban",
                    values=c("1"="navyblue", "2"="orange", "3"="darkgreen"))
```



Modified:

```
## Warning: A numeric 'legend.position' argument in 'theme()' was deprecated in ggplot2
## 3.5.0.
## i Please use the 'legend.position.inside' argument of 'theme()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

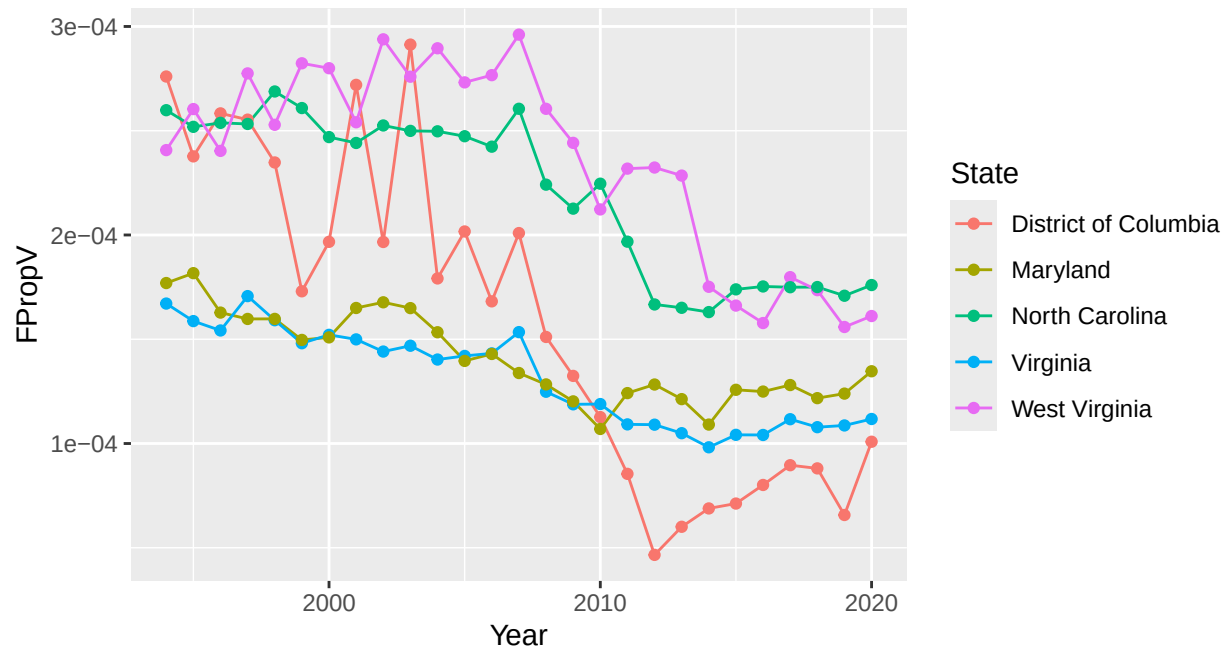


Problem 3

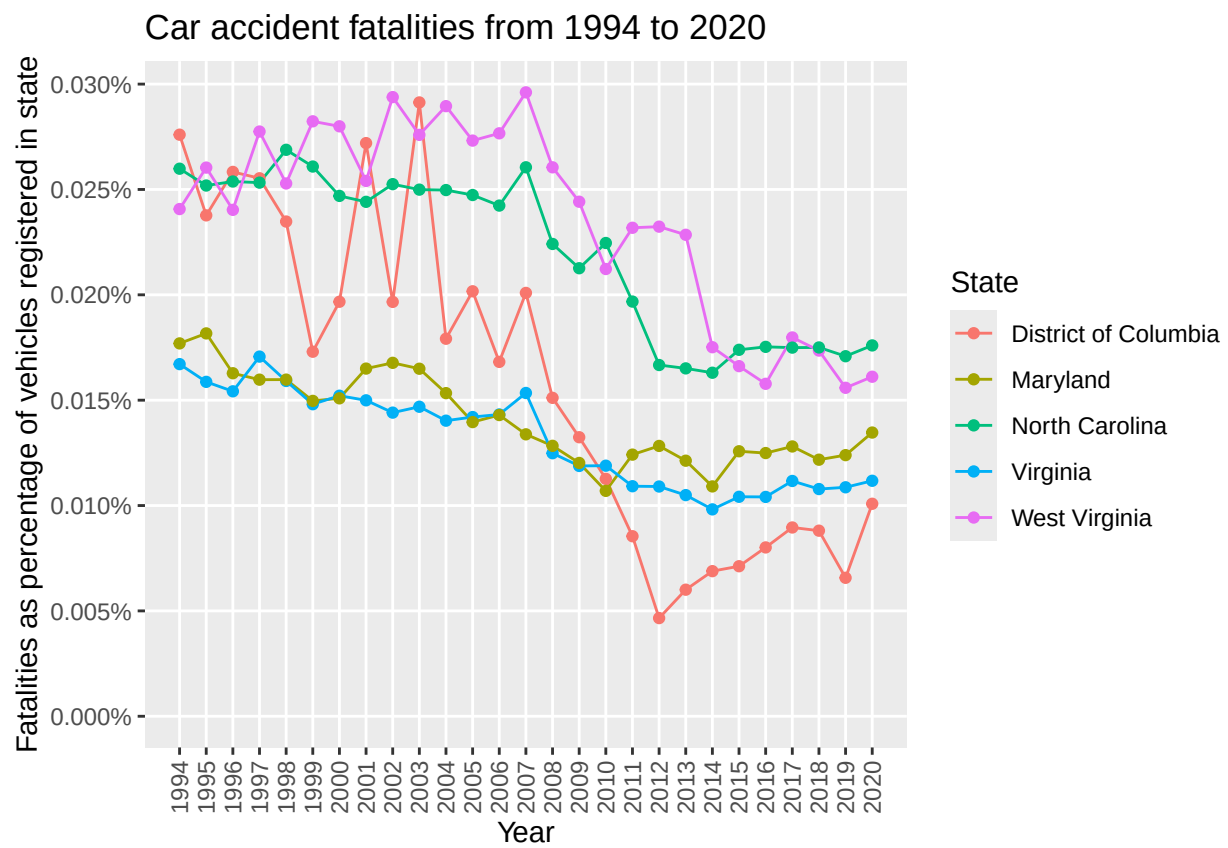
This plot is from Homework 5 problem 2e:

```
fatal.df <- read.csv("fatalities.csv",
                     header=TRUE)
fatal.df$FPropV <-
  fatal.df$Fatalities/(fatal.df$Registered.Vehicles*1000)

ggplot(fatal.df, aes(x=Year, y=FPropV, color=State)) +
  geom_line() + geom_point()
```



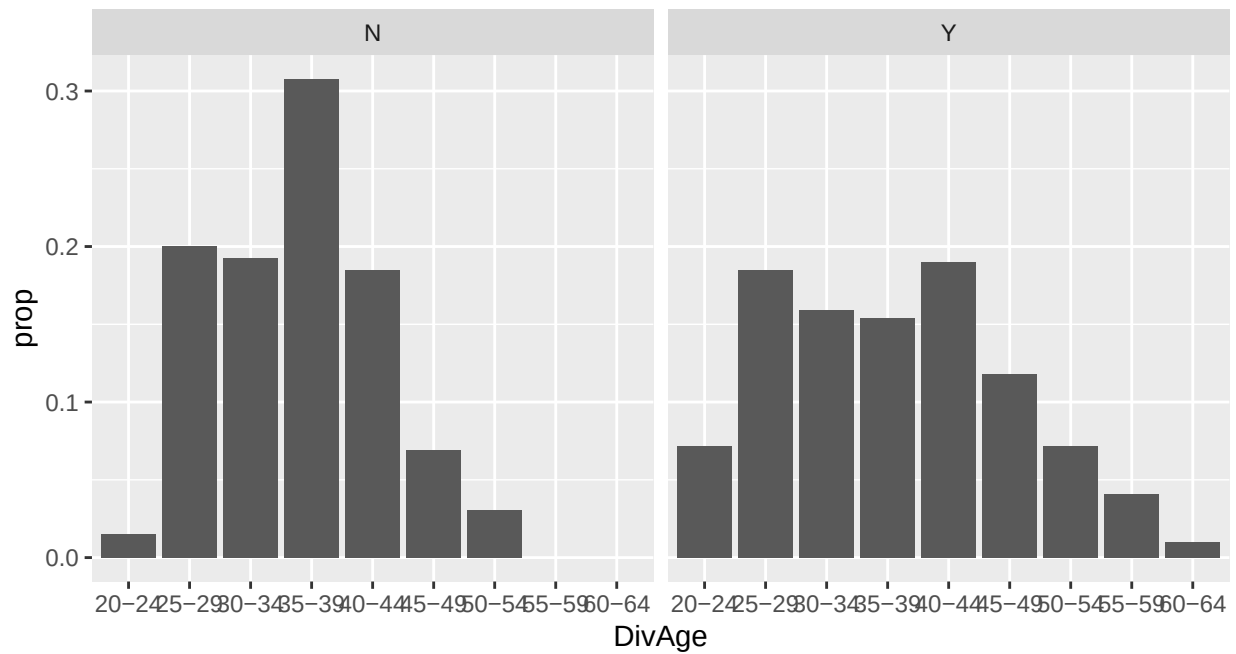
Modified:



Problem 4

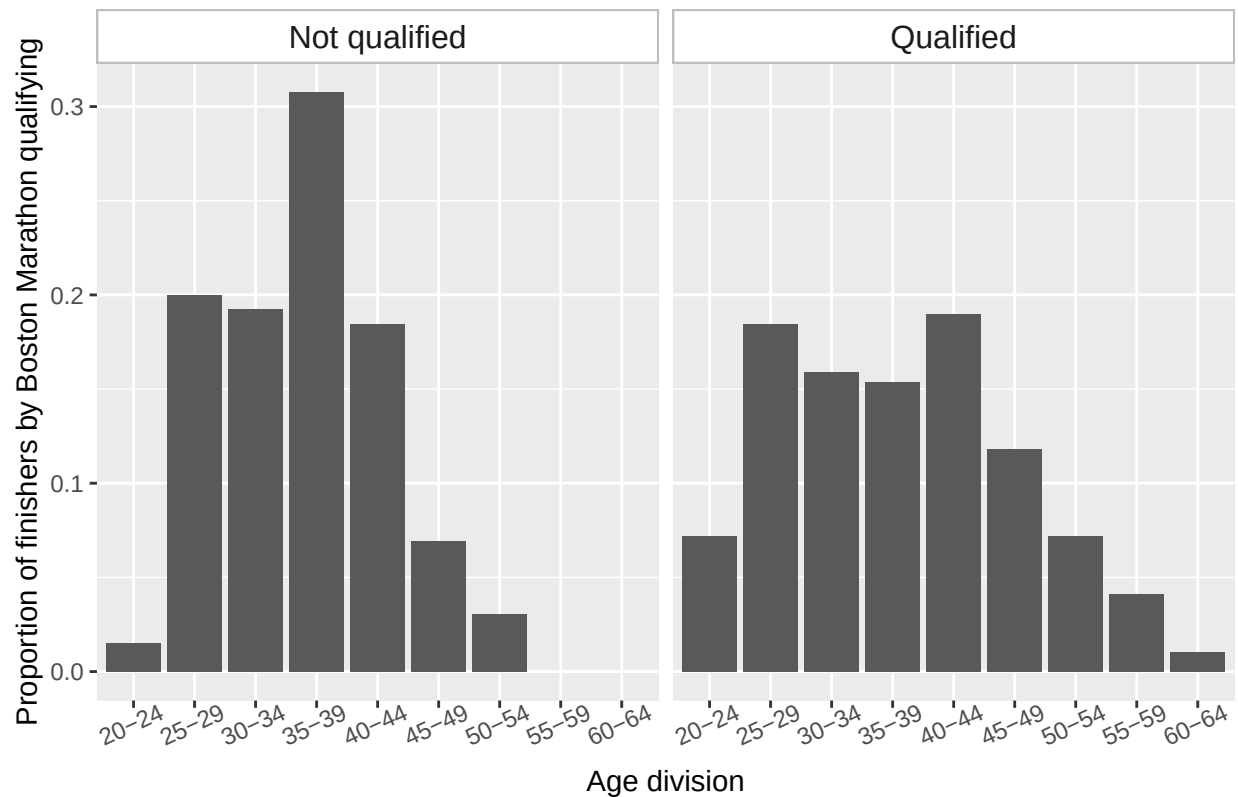
Graphic shown in problem 1e above:

```
ggplot(nym21, aes(x=DivAge, group=BostonQualifier)) + geom_bar(aes(y=after_stat(prop))) +  
  facet_grid(~BostonQualifier)
```



Modified:

Select 2021 NYC Marathon finishers

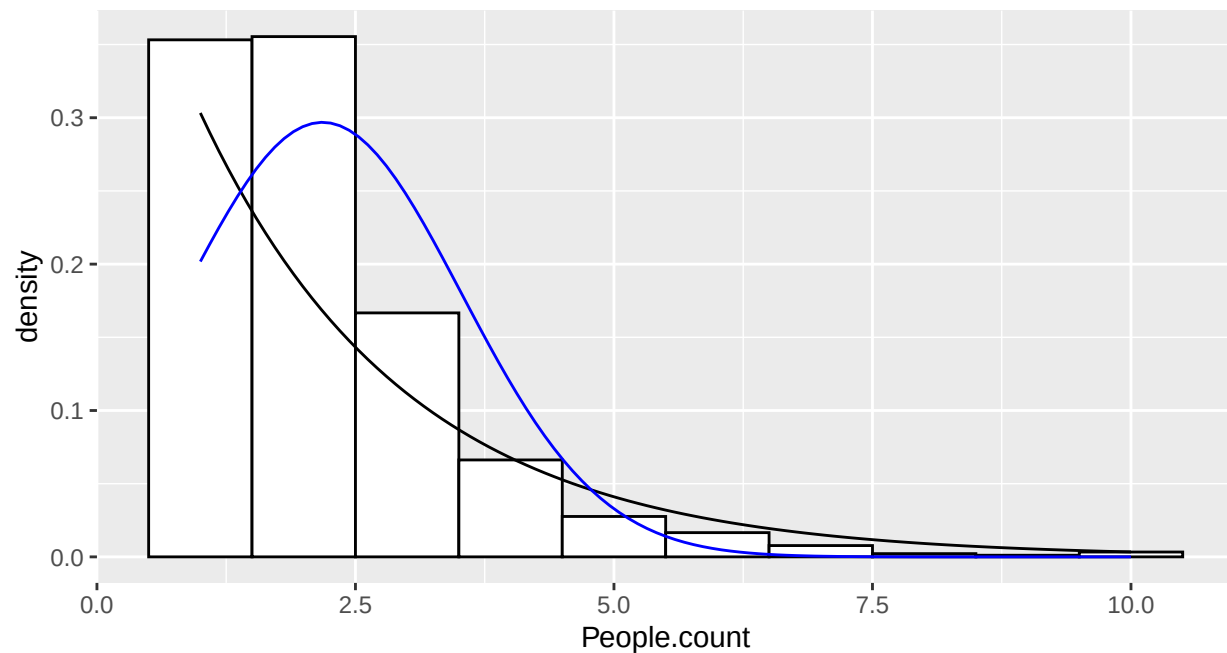


Problem 5

New graphic using the **Virginia** information in the file `fatal_accidents.csv`:

Note: Both curves are common statistical distributions that use sample statistics for parameters where appropriate.

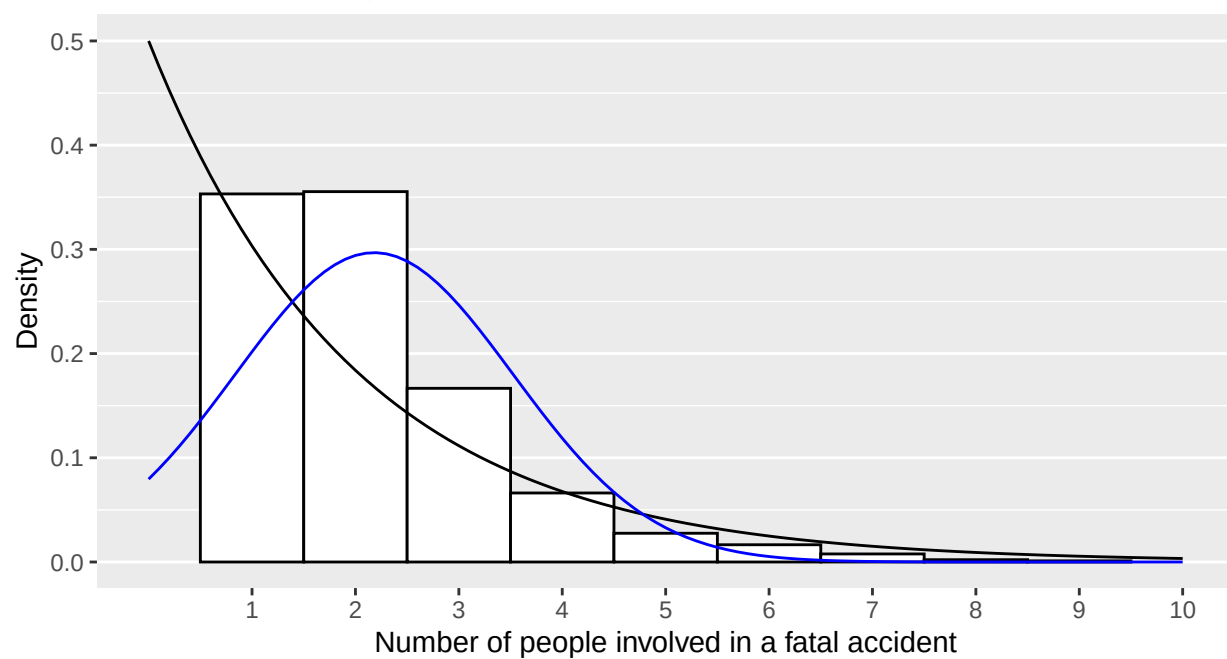
```
fatal <- read.csv("fatal_accidents.csv",
                 header=TRUE)
fatal.va <- fatal[fatal$State=="Virginia",]
fatal.va$People.count <- fatal.va$People.count.IN + fatal.va$People.count.OUT
mean.p <- mean(fatal.va$People.count)
sd.p <- sd(fatal.va$People.count)
ggplot(fatal.va, aes(x=People.count)) +
  geom_histogram(aes(y=..density..), binwidth=1, color="black", fill="white") +
  stat_function(fun=dchisq, args=list(df=2)) +
  stat_function(fun=dnorm, args=list(mean=mean.p, sd=sd.p), color="blue")
```



Modified:

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_bar()').
```

Distribution of Virginians involved in fatal accidents in 2020



Problem 6

What conclusions can you draw about the distribution of the number of people involved in fatal accidents in Virginia in 2020?

Type your response in the appropriate Gradescope free response box.